ARXIS

AEC Architecture Traceability & Engineering Change Governance

Graph-Theoretic Dependency Management for Complex Built-Asset Infrastructure

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Live Case Study | GitHub Repository
Standards: ISO 19650 | FIDIC / AIA

Executive Summary & Strategic Context

In large-scale Architecture, Engineering, and Construction (AEC) capital projects, uncoordinated design revisions represent the single greatest cause of budget overruns and contractor delay claims. On-site physical rework accounts for 5% to 12% of total project expenditure, frequently triggered when a localized architectural change invalidates structural load limits or MEP (Mechanical, Electrical, Plumbing) routings without cross-disciplinary synchronization.

ARXIS was engineered at **Arxa Group** to establish structural dependency integrity across capital facilities. Operating under the institutional axiom: *traceability does not grant approval*, ARXIS models requirements, engineering decisions, and digital deliverables as a formal **Directed Acyclic Graph (DAG)**. ARXIS automatically traces the topological blast radius of any design change, locks deliverables to cryptographic version hashes, and mandates explicit RACI sign-offs before physical construction begins.

System Architecture & Graph Mechanics

The platform coordinates capital project information flows across four mathematical tiers:

1. **Tier 1: Client Requirements:** Formalizes functional, environmental (LEED), and regulatory mandates into unique nodes (REQ_ID). Establishes unambiguous root intent.
2. **Tier 2: Engineering Decisions:** Maps trade-off decisions made by lead engineers across disciplines (DEC_ID). Links choices directly to governing requirement IDs and registered approval authorities.
3. **Tier 3: Design Deliverables (Artifacts):** Ingests BIM models (Revit / IFC), 2D CAD sheets, and technical schedules (ART_ID). Binds each deliverable to its upstream decision and SHA-256 content hash. Strictly eliminates orphan deliverables.
4. **Tier 4: Topological Change Impact Engine:** When an Engineering Change Request (ECR) is submitted, the engine uses Kahn's algorithm and depth-first traversal to compute the reachable impacted subgraph $\mathcal{R}(u)$. All downstream documents are automatically tagged as `PENDING_REEVALUATION`, preventing premature site execution.
5. **Tier 5: RACI Authorization Gates:** Enforces role-based sign-offs. Rejects baseline releases unless certified discipline leads (Architectural, Structural, MEP) have approved all linked decisions.

5-Dimensional Multidimensional Operating Model

In accordance with Arxa Group Systems Architecture Standards, ARXIS integrates five operational dimensions:

- **Technical Architecture:** NetworkX/Python graph core, strict cycle-detection algorithms, OpenAPI 3.1 REST contracts, and immutable JSON-schema manifests.
- **Operating Workflow (RACI):** Project Director (authorizes change budgets) $\rightarrow$ Discipline Leads (verify technical feasibility) $\rightarrow$ Document Controller (locks Common Data Environment baselines) $\rightarrow$ General Contractor (verifies field installation against locked version hashes).
- **Financial Model & Governance Economics:** Evaluates structured change coordination effort (€250–€490 per change event) against catastrophic physical clash rework expenses (€50,000–€250,000 per site clash). Demonstrates a **Return on Governance exceeding 200× to 1000×**.
- **Risk & Governance Model:** Complies with ISO 19650 (BIM information containers and state transitions) and international construction dispute standards (FIDIC Red/Yellow Books, AIA Document G701).
- **Scaling & Integration:** Native IFC 4.3 open schema adapters, direct Common Data Environment (CDE) webhooks (Autodesk Construction Cloud, Procore).

Empirical Verification & Benchmark Performance

The reference implementation is verified via 7 automated regression test suites in the public repository (arxis-traceability on GitHub):

Benchmark Metric	Observed Score	Traditional Baseline	Operational Significance
Graph Dependency Coverage	**100.0%**	60.0–75.0%	Deterministic verification: zero orphan decisions or deliverables.
Cycle Detection Latency	$<$ **5 ms**	Manual / Days	Kahn's algorithm instantly detects and halts circular dependencies.
Blast Radius Traversal	**100.0%**	Subjective	Transitive closure tags every downstream affected drawing.
Unauthorized Actor Blocks	**100.0%**	0.0% (Informal)	Prevents unauthorized verbal approvals; logs security faults.
Average Clash Avoidance ROI	$>$ **200×**	1.0×	A €250 review prevents €50k+ in site emergency refabrication.

Architecture Decision Records (ADRs) & API Contracts

All architectural trade-offs are formally governed by RFC-style Architecture Decision Records:

- **ADR-0001:** Directed Acyclic Graph (DAG) for Change Blast Radius Traversal.
- **ADR-0002:** ISO 19650 Common Data Environment (CDE) Deliverable Locking & Hash Manifests.
- **ADR-0003:** Programmatic RACI Gate Enforcement Prior to Construction Baseline Release.
- **OpenAPI 3.1 Contract:** Endpoints for '/v1/change-requests/submit', '/v1/impact/blast-radius', and '/v1/gates/sign-off'.

Methodological & Confidentiality Notice: This architectural whitepaper utilizes validated synthetic benchmark datasets and open institutional AEC standards (ISO 19650, IFC 4.3, FIDIC). No confidential corporate banking or private client data from active or past institutional engagements (BNP Paribas CIB, Santander CIB) is contained herein. © 2026 Igor Grybek. All rights reserved.